en(B)

Twenty-second International Linguistics Olympiad

Taipei (Taiwan), 20–27 July 2025

Individual Contest Problems

Rules for writing out the solutions

Do not copy the problems. Write down your solution to each problem on a separate sheet or sheets. On each sheet indicate the number of the problem, the number of your seat and your surname. If you do not do this, your work may be mislaid or misattributed.

Unless stated differently, you should describe any patterns or rules that you identified in the data. Otherwise your solution will not be awarded full marks.

Problem 1 (20 points). Here are some numbers in Dzongkha and their numerical values:

1 — **ci** 3 — **sum** 8 — **ge** 12 — **cuɲi** 17 — **cupdyn** 19 — **cygu**

For higher numbers, Dzongkha uses two different systems (referred to here as A and B). Below are some numbers written in both systems, as well as their numerical values:

System A	System B	Value
ke ci da ɲi	**tsaɲi**	22
ke ci da ŋa	**tsaŋa**	25
ke pɟe-da ɲi	**sumcu**	30
ke ci da cyʑi	**soʑi**	34
ke ɲi da dyn	**ʑedyn**	47

System A	System B	Value
ke ko-da sum	**ŋaŋa**	55
ke sum da cuɖu	**døndu**	76
ke ʑi	**gepcu**	80
ke ʑi da gu	**ɟagu**	89
ke ceŋa	**sumɟa**	300

Finally, some equalities are given with left-hand side written in system A and right-hand side written in system B. Some numbers are missing.

	System A		System B
(1)	**cusum + ke pɟe-da ʑi**	=	**ɟasum**
(2)	**ɲiɕu ɲi**	=	**ɲiɕu × ʑipcu**
(3)	**ɲiɕu ci da ke sum da gu**	=	**(ŋapcu × gu) + cygu**
(4)	**ɲiɕu pɟe-da ɲi + ke pɟe-da ɖu**	=	**ŋapɟa + ɲiɟa cutãm**
(5)	**(ɲi × ko) + pɟe**	=	**ɲi**
(6)	**(ɲiɕu ko-da sum × pɟe) + ke pɟe-da sum**	=	**ɖukɟa**
(7)	**ɲiɕu ci da ke cuɖu da cuɖu**	=	**(ɟaʑi × ʑi) + ʑipɟa**
(8)	**ɲi × ɲiɕu ci da ke cutãm da gu**	=	**(____X × ɲiɕu) + copge**
(9)	____Y **+ ke ci da ʑi**	=	**ɟaɖu**
(10)	____Z **+ ke ko-da ɖu**	=	**dynɟa + sumɟa**

(a) Fill in the blanks X–Z with Dzongkha numbers.

(b) Write with digits the equalities (1–10).

(c) Write in Dzongkha in both systems: 75; 570.

⚠ The Dzongkha language belongs to the Sino-Tibetan family. It is spoken by approx. 171,000 people in Bhutan.

The words are given in a simplified transcription. **ɖ**, **ɟ**, **ɲ**, **ŋ**, **ɕ** and **ʑ** are consonants. **ã**, **ø** and **y** are vowels. *—Vlad A. Neacşu*